An EV Charging Station Placement for Ride-hailing Service: from Big Data Networking Perspective

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***Abstract*—The charging station is an important supporting infrastructure for electric vehicles (EVs), and its placement will directly affect the development of the whole industry chain of EVs. Although many studies have investigated the placement of EV from the distributions of EVs, few works investigate the charging station placement in a ride-hailing service provider perspective. As a ride-hailing service provider, it does not only provide the ride-hailing service, but also the power supply for hailing vehicles for the purpose of cost saving. In this paper, the charging station placement for a ride-hailing service provider has been studied from networking perspective. Inspired by the cache placement problem in content distribution networking (CDN), the charging station is analogous to the cache server, and its placement is designed according to the access frequency and the demand for content, namely, electricity, aiming at enabling more users to complete the charging behavior nearby. Both hotspot path and EVs' energy demand and consumption are considered to tackle the placement of charging stations. Finally, we use hit rate and distance cost as metrics to evaluate the performance of placement of charging stations. The evaluation results confirm the superiority and feasibility of the proposed scheme.**

I. Introduction

Ride-hailing has become a proliferating service in the shared economy era. Meanwhile, the penetration rate of electric vehicles (EVs) has been increasing sharply due to the continuous innovation of battery technology. So the use of EVs in ride-hailing service is becoming a future trend [1]. However, EV charging stations placement is far from meeting the normal charging needs of EVs' users, not to the charging of EVs for ride-hailing service. Therefore, for ride-hailing service providers, it is necessary to build their own EV charging stations for hailing vehicles. The problem of charging station placement is inevitable and could be costly if done improperly.

Aiming at the problem of EVs charging stations, several researches have studied method for charging station placement. Saeed Shojaabadi et al. [2] proposed a mathematical model by considering the rate of customer participation in demand response plan. Y. Li et al. [3] used a large number of electric taxi trajectory data for analysis and developed an optimal charging station deployment framework. Q. Yan et al. [4] proposed a four-stage intelligent optimal control algorithm which combined with commercial buildings. J. Tan et al. [5] proposed a new bi-level charging station planning method for distribution and transportation integrated system through considering the influence of the operation strategy of electric vehicle charging station on distribution system and traffic system. K. Qin et al. [6] extracted the hotspots of the city through taxi trajectory data, mainly introducing the cluster analysis technology of the data field into the extraction and descent points of the taxi data, and using time weights to improve method. Y. Dong et al. [7] used an improved K-means clustering algorithm to divide the urban area into several service areas, and then used the most location model to calculate the most location of the charging station. These previous studies mainly focus on the placement of EV charging stations from uniform traffic flow while neglecting the requirement of certain types of service, such as ride-hailing. And the characteristics of urban EV and the needs of passengers have not been fully considered.

Inspired by the big data networking, we find that there is a high similarity between the idea of laying out cache server in the content distribution network (CDN) and laying out charging stations for ride-hailing service. The key of CDN is to place a large number of caching servers in areas where users have relatively concentrated access in the network. The purpose is to improve the speed of users' access to the website and alleviate the situation of network congestion. The placement of EV charging stations for ride-hailing service is similar to that of cache server in CDN, which the placement of EV charging stations needs to be accessed in high frequency to reduce the cost of the placement and meanwhile satisfy EVs' power supply requirement. For this purpose, placement of charging station, namely, the placement of "cache server" in the road network, we consider the hotspot path where charging stations are visited in high frequency, the high power consumption area and high power demand intensity area.

In this paper, we first study and analyze ride-hailing service data from DiDi ride-hailing service that can fully reflect ride-hailing service requests of residents and the distribution of travel demand [8]. We then formulate the EV charging station placement for ride-hailing service as a problem to strike a balance between user demands and construction costs [9]. As for target placement location mining, we use clustering algorithm based on grid density to find hotspot path. Meanwhile, considering the endurance of EVs, we analyze the characteristics of driving dynamics and batteries, calculate the energy consumption of EVs, and extract the spatial distribu-

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tion of high energy consumption areas and charging demand intensity areas. By combining both factors, the Ride-Hailing Charging Station Placement (RHCSP) approach is proposed. The evaluation results reveal that the proposed EV station charging placement approach can meet the requirement of ride-hailing service.

II. Preliminaries

To show the placement of EV charging station for ride-hailing service, the data trace from a ride-hailing service company and basic approach is illustrated in this section.

A. Dataset

The ride-hailing service data is collected from DiDi company, covering the area within the third ring road of Chengdu city with an area of 77 km^2. The data trace contains 20,000 vehicles, and it recorded the trajectory of each vehicle by GPS in a grain step of 2-4 seconds. In order to view the coverage status of dataset more intuitively, we import the shapefiles of Chengdu road network into visualization software, ArcGIS 10.2. The obtained visual map of trajectory data is shown in Fig. 1. It shows that the hailing-vehicles mainly distribute in the urban area.

Fig. 1. The reconstruction of the trajectory

B. Basic Approach

Considering that cache server in CDN is deployed on the basis of content with high user access frequency and high demand, as a "cache server" in urban road network, EV charging station also need to consider to charge the "content" with high user access frequency and high demand. The area with high driving frequency is similar to the "content" with high user access frequency, while the area with high power consumption of EVs and high power demand intensity are similar to the content with high demand. Therefore, to solve the problem of EVs charging station placement, we are confronted with three principal questions: (1) How to find out the area with high driving frequency? (2) How to identify high power consumption areas? (3) How to identify high power demand intensity area? For the first question, we introduce hotspot path to represent the high driving frequency area. Then we mine the hotspot path. For the second question, we measure the consumption of vehicles and find the distribution of power consumption in the area to extract the location of high power consumption. For the third question, we consider the areas where EVs are about to run out of electricity as the areas with high power demand intensity. By combining these factors, a ride-hailing charging station placement approach, named RHCSP, is proposed.

III. Hotspot path algorithm

The definition of the hotspot path is the section of the road with high driving frequency, which is mined by the spatial distribution characteristics of the vehicle's trajectories. In order to obtain the distribution of hotspot paths, we mainly use the spatial application noise spatial clustering algorithm based on grid density. This method combines the gridding method with density-based clustering algorithm (DBSCAN).

A. Gridding Method

The gridding method is to quantize the space into many grids and then cluster on these grid cell, as shown in Fig. 2, the geographical area is divided into many blocks.

Each grid has its latitude and longitude range, and each set of GPS data is mapped to its corresponding grid according to its latitude and longitude. By gridding the area, algorithm does not operate on each individual data point, which greatly reduces the size of the data to be processed and improves the calculation efficiency.

Fig. 2. Gridding Method

B. The hotspot path mining

Dataset is the spatial data points carrying latitude and longitude coordinates, $S_P = \{p_1, p_2, p_3, p_4, \ldots, p_n\}$, each point $p_i = (lon_i, lat_i)$. Then we divide the area into several $\mathbf{k} \times \mathbf{k}$ grids. This area covered by the spatial dataset A, $\mathrm{A} = [\mathrm{lon}_{\min}, \mathrm{lon}_{\max}] \times [\mathrm{lat}_{\min}, lat_{\max}]$. The size of each grid is $\mathbf{k}$. Using mapping relation to map all data points into corresponding grids. Then these data will be included in a grid cell, the index of its grid is $ind_{\mathrm{lon}} = \left[\frac{\mathrm{lon}_p - \mathrm{lon}_{\min}}{k}\right]$, $ind_{\mathrm{lat}} = \left[\frac{\mathrm{lat}_p - \mathrm{lat}_{\min}}{k}\right]$. After that we calculate the grid density and the coordinates of all grids. Grid density is the number of all data points mapped into this grid, given a grid cell G_i, its grid density denoted as $\mathrm{den}(G_i)$. And grid coordinates means that the average latitude and longitude of all the data points in the grid , $\mathrm{lon}(G_i) = \frac{\sum_{i=1}^{n} \mathrm{lon}(p_i)}{n}$, $\mathrm{lat}(G_i) = \frac{\sum_{i=1}^{n} \mathrm{lat}(p_i)}{n}$. Then obtain the set of grids. When the gird density is greater than a certain value, $\mathrm{den}(G_i) > \gamma$, that is hotspot grid. Judge the directly reachable grid set and reachable grid set according to the calculation of the spatial distance between the grids. The definition of directly reachable grid is that for a set of grids S_G, if $\exists G_i, \quad G_j \in S_G, \mathrm{dis}(G_i, G_j) \leq dis_{min}$, then call G_i, G_j directly reachable grid, dis_{min} is distance threshold. The definition of reachable grid is that if there a grid cell sequence $G_1, G_2, G_3, \ldots, G_{n-1}, G_n$, where $G_1 = Gi$, $G_n = Gj$, $1 \leq r < n$, G_r is directly reachable from G_{r+1}, so G_i is G_j reachable grid. And the definition of spatial distance is the spherical distance between two points on the surface of the earth $\mathrm{dis}(p_A, p_B)$. For a subset H of a non-empty grid set S_G, the area covered by this set is the hotspot area if it satisfies the following three conditions: 1. Denseness: $\forall G_i \in \mathrm{H}, G_i$ is the hotspot grid; 2. Connectivity: $\forall G_i, \quad G_j \in \mathrm{H}, \quad G_i$ is reachable grid of G_j; 3. Maximality: $\forall G_j \in \mathrm{H}$, if G_i is directly reachable grid of G_j, then $G_i \in$ H. So according to this algorithm, we can obtain the set of the final hotspot areas.

IV. Electric vehicle energy calculation

In CDN system, the best caching server node is mined according to network traffic and load condition. In order to obatin the best placement of charging station, "cache" in road network, we extract the distribution of EVs power consumption on the road and the distribution of EVs power denmand intensity.

A. Power consumption calculation

For an EVs, the battery output power is closely related to the driving state of the vehicle, that is, the battery pack' output power must satisfy the required power during the running of the vehicle. According to the dynamic analysis of the vehicle during driving, when an EV is running with a velocity, $\mathrm{v}[m/s]$, and an acceleration, $\mathrm{a}[m/s^2]$, then the driving resistance of the vehicle is the sum of acceleration resistance, $F_a = ma$, the rolling resistance, $F_r = mg\mu$, the air resistance, $F_k = kv^2$, and the inclination resistance, $F_g = mg\sin\alpha$. According to the power calculation formula, $\mathrm{P} = \mathrm{F} \cdot \mathrm{v}$, then combining the coefficient obtained by the actual situation of the road driving of the vehicle, we can obtain $P = \frac{v}{1000\eta}\left(G\mu\cos\alpha + \frac{1}{2}\rho_a \cdot C_d \cdot A_f \cdot v^2 + G \cdot \sin\alpha + m \cdot \delta \cdot a\right)$, where V is the instantaneous of the vehicle, m/s, η is the transmission efficiency of the power unit to the driver wheel, G is the weight of the vehicle, N, A is the inclination angle of the road surface, ρ_a is the air density, g/L, c_d is the air resistance coefficient determined according to the shape of the vehicle body, A_f is the windward area of the vehicle, m^2, M is the mass of vehicle, kg, δ is the moment of inertia, a is the instantaneous acceleration of the vehicle, m/a^2, μ is the rolling resistance coefficient.

So when the vehicle is driving on a flat road, that is, $\alpha = 0$. The electric vehicle power is expressed as: $P = \frac{v}{1000\eta}\left(G\mu + \frac{1}{2}\rho_a \cdot C_d \cdot A_f \cdot v^2 + m \cdot \delta \cdot a\right)$. For each trip of EVs, its trajectory data is composed of several data points including longitude and latitude, $S_P = \{p_1, p_2, p_3, p_4, \ldots, p_n\}$, each point $p_i = (lon_i, lat_i)$. we divide each trip into several sections, so we can obatin $S_P = \{d_1, d_2, d_3, d_4, \ldots, d_{n-1}\}$, $d_i = (lon_i, lat_i; lon_{i+1}, lat_{i+1})$. We extract velocity and acceleration according to the distance and time difference between two points. So the power consumption of every journey is $P(S_p) = \sum_{i=1}^{n-1} P_{d_i}$, P_{high} is high power consumption threshold. When $\sum_{i=1}^{m} P_{d_i} \geq P_{high}$, $m \in [1, n-1]$, we think that $d_i = (lon_i, lat_i; lon_{i+1}, lat_{i+1})$ is the section where the vehicle urgently needs to be charged [10].

B. Power demand intensity calculation

In order to obtain the distribution of power demand intensity, we calculate the location of electric vehicles in urgent need of power supply during each trip. The battery effective energy expression for an EV is: $E = V_p \cdot Q_0 \cdot s \cdot \eta_{DOD} \cdot \eta_{dis}$, where V_p is the terminal voltage of the battery pack, $Q_0$is the rated capacity of the battery pack, s is the value of state of charge (SOC), and η_{DOD} is the battery discharge depth, η_{dis} is the battery discharge efficiency. Electric vehicle discharge time is: $t = \frac{E}{p}$, the driving range of EVs is: $S = v \cdot t$, the electric energy calculation formula for EVs is: $e = \frac{E}{S} = \frac{P}{v}$. Because each trip of EVs can be expressed as: $S_P = \{d_1, d_2, d_3, d_4, \ldots, d_{n-1}\}$, $d_i = (lon_i, lat_i; lon_{i+1}, lat_{i+1})$, the eletric energy consumption for every journey is $e(S_p) = \sum_{i=1}^{n-1} e_{d_i}$. Assume that e_{all} is the total electric energy value of an electric vehicle, α is the power demand intensity factor. When $\sum_{i=1}^{m} e_{d_i} \geq \alpha \cdot e_{all}$, $m \in [1, n-1]$, we think that $d_i = (lon_i, lat_i; lon_{i+1}, lat_{i+1})$ is the section where the vehicle urgently needs to be charged.

V. RHCSP Scheme

RHCSP scheme is based on hotspot path mining and EVs energy calculation. This scheme takes into account the related factors of road access frequency, high power consumption and high power demand intensity. The specific steps of RHCSP scheme are as follows:

- Step1:
  Hotspot path mining algorithm takes road access heat as index to extract target location, these target locations are several cluster coordinates of hotspot paths, $S_{hotspot} = \{S_{G_1}, S_{G_2}, S_{G_3}, S_{G_4}, \ldots, S_{G_n}\}$, $S_{G_a} = \{G_{a,1}, G_{a,2}, G_{a,3}, G_{a,4}, \ldots, G_{a,b}\}$ is hotspot cluster, $a \in [1, n]$. And, $G_{a,m} = \{p_{m,1}, p_{m,2}, p_{m,3}, p_{m,4}, \ldots, p_{m,j}\}$ is hotspot grid, $m \in [1, b]$, $p_{m,i} = (lon_{p_i}, lat_{p_i})$ is the coordinate of point, $i \in [1, j]$. Firstly, we find out the hotspot cluster with the least data points and take it as a unit cluster. The number of charging stations in this cluster is 1. Based on this method, we deduce the

number of charging stations in all other non-unit clusters. Then, we calculate the coordinates of charging stations according to the coordinates of clusters, $p\left(S_{G_a}\right) = \left(\frac{\sum_{m=1}^{b}\sum_{i=1}^{j} lonp_{m,i}}{b \cdot j}, \frac{\sum_{m=1}^{b}\sum_{i=1}^{j} latp_{m,i}}{b \cdot j}\right)$, so the final coordinates set is $p_{hotspot} = \{p\left(S_{G_1}\right), p\left(S_{G_2}\right), p\left(S_{G_3}\right), p\left(S_{G_4}\right), \ldots, p\left(S_{G_n}\right)\}$.

- Step2:
  By calculating the power consumption of EVs introduced in Section IV, we extract the coordinates of all sections where power consumption is higher than the power consumption threshold P_{high}, that is, the set of high power consumption areas, $p_{high-power} = \{p_{d_1}, p_{d_2}, p_{d_3}, p_{d_4}, \ldots, p_{d_n}\}$, p_{d_i} is the coordinates of the section d_i with high power consunmption. And by calculating the power demand intensity, we obtain the coordinates of the areas in urgent need of charging, $p_{high-demand} = \{p_{d_1}, p_{d_2}, p_{d_3}, p_{d_4}, \ldots, p_{d_n}\}$, that is, the areas with high power consumption demand intensity, p_{d_i} is the coordinates of the section d_i with high power demand intensity.
- Step3:
  Take the union of these three sets, that is, $p_{union} = p_{hotspot} \bigcup p_{high-power} \bigcup p_{high-demand}$, calculate the distance between all data points in p_{RHCSP}. When the distance is less than the distance threshold dis_{union}, we think that these two points belong to the same coordinate points, then delete one of them. Finally, the coordinate point set of the charging station site of electric vehicle p_{RHCSP} is obtained.

VI. Evaluation Experiment

In this section, we set up evaluation experiments to show the performance of charging station placement schemes.

A. Experiment setup

We use ride-hailing service data as described in Section II for the experiment. Among them, 70% of the data are used for EVs charging station placement scheme and 30% for comparative experiment, which mainly analyses and evaluates the performance of RHCSP scheme and other baseline schemes. For the experiment, we assume that the electricity of the each vehicle is in a full state before it starts its first trip every day. And in the part of hotspot path mining, we use the distance threshold $dis_{min} = 500$, the minimum number of data points of the hot grid $\gamma = 420$. In the part of EVs energy calculation, we use high power consumption threshold $P_{high} = 30$, EVs total electric energy value $e_{all} = 50$, power demand intensity factor $\alpha = 80\%$, when the electricity of a vehicle elapsed 80 %, the vehicle is considered as high power demand, the weight of EV is $m = 1500\,\mathrm{kg}$, transmission efficiency η =96%, rolling drag coefficientμ =0.005, air density ρ_a =$1.293\,\mathrm{g/L}$, air resistance coefficient $c_d = 0.35$, windward area $A_f = 2\,\mathrm{m}^2$ and moment of inertia $\delta = 1.2$.

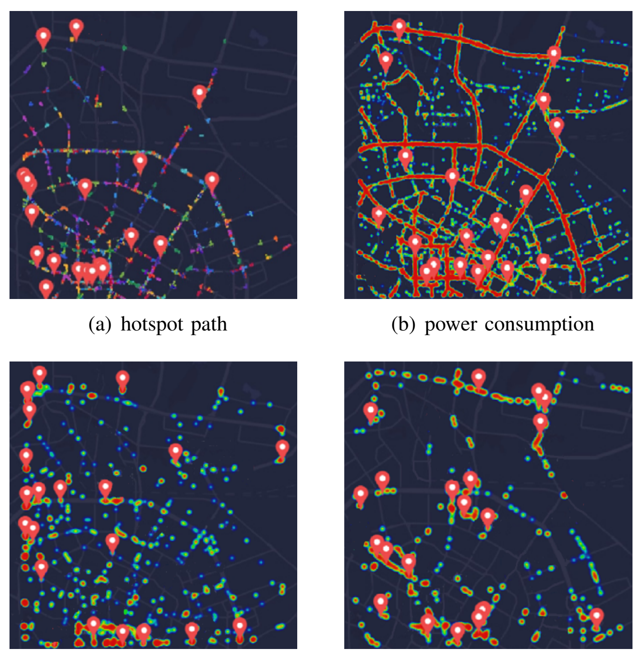
In the part of comparative experiment, we test the performance of four placement schemes: Hotspot Path [11], Power Consumption [12], Demand Intensity [13] and RHCSP. Firstly, the same experiment data are processed by these four different methods, and then obtain the four placement results. Next, the same number(n_{cs}) of charging stations are selected in these four placement results to ensure that the next test is based on the same construction cost. To evaluate the performance of schemes, the two metrics are measured: hit rate and distance cost.

- Hit rate: The ratio of the vehicle successful reaching the charging station before the energy is exhausted and the total vehicle.
- Distance cost: The average distance between the place where the vehicle starts to drive to charging station and the charging station.

Finally, the test data are used to calculate how many vehicles can successfully reach the location of a charging station within the range of their endurance, and the distance between the location when the power consumption elapses 80% and charging station is recorded .

B. Evaluation results

This section shows the results of comparative experiments. As shown in Fig. 3 below, this is the result of four placement schemes when the number of charging stations $n_{cs} = 20$.

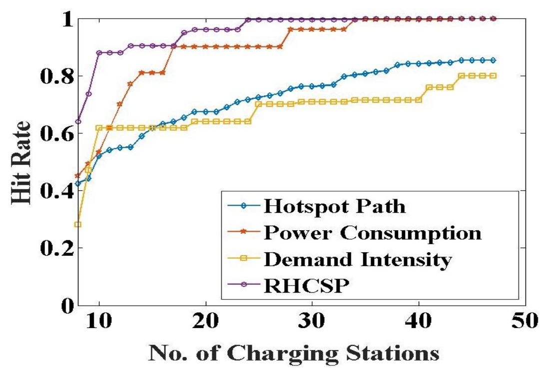


(a) hotspot path
(b) power consumption
(c) demand intensity
(d) RHCSP

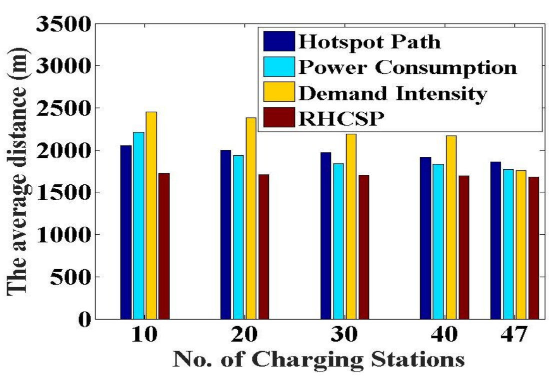
Fig. 3. The placement results of schemes when the number of charging station is 20

Fig. 3(a) shows the distribution of charging station based on hotspot path. The red sign in the picture represents the location of the charging station, and the different colors on the road represent different hotspot path. This scheme chooses the location according to the frequency of road access. The closer to the downtown area, the higher the traffic volume. Therefore, the more hot spots there are. Fig. 3(b) shows the distribution of charging station based on high power consumption. The red color in the figure indicates the higher power consuming section, while the green color indicates the lower power consuming section. The distribution of charging stations in this scheme is mainly on the main roads and concentrated on the sections in the urban center. Fig. 3(c) shows the distribution of charging station based on demand intensity. The station locations in this scheme are mainly on the main roads of one ring, two rings and three rings, which are distributed on the sections in the marginal areas. Fig. 3(d) shows the distribution of charging station based on RHCSP. The site distribution of this scheme is average, and several relatively centralized locations are mostly railway stations, important transportation hubs and central business districts.

According to the comparison in Fig. 4, RHCSP shows the highest hit rate. When the number of charging stations equals 20, the hit rate of RHCSP is 47% higher than that of Hotspot Path, 11% higher than that of Power Consumption and 56% higher than that of Demand Intensity.



(a) Charging station hit rate



(b) Charging station distance cost

Fig. 4. The comparative results

Distance cost reflects the convenience of the driver to reach the charging station. The comparison results show that the average distance cost decreases with the number of charging stations increase. When the number of charging stations equals 20, the hit rate of RHCSP is 14.5% higher than that of Hotspot Path, 11.8% higher than that of Power Consumption and 28.3% higher than that of Demand Intensity..

The results of the performance evaluation show that RHCSP has the highest hit rate and the lowest distance cost under the limited construction cost. So this scheme has the best performance in striking a balance between cost and user satisfaction.

VII. Conclusion

In this paper, we proposed RHCSP for charging stations placement of EVs for ride-hailing service from Big Data Networking Perspective, which combines hotspot path, areas with high power consumption and high demand intensity. Hotspot path ensures the visiting frequency of charging stations, mining high energy consumption areas and areas that are about to run out of electricity can obtain the distribution of urgently needed energy supply in the road network. The evaluation results show that RHCSP scheme has the highest hit rate and lowest distance cost.

References

[1] X. Wang, C. Yuen, N. U. Hassan, N. An, and W. Wu, "Electric vehicle charging station placement for urban public bus systems," *IEEE Transactions on Intelligent Transportation Systems*, no. 99, pp. 1–12, 2017.

[2] S. Shojaabadi, S. Abapour, M. Abapour, and A. Nahavandi, "Optimal planning of plug-in hybrid electric vehicle charging station in distribution network considering demand response programs and uncertainties," *Iet Generation Transmission & Distribution*, vol. 10, no. 13, pp. 3330–3340, 2016.

[3] Y. Li, J. Luo, C. Y. Chow, K. L. Chan, D. Ye, and Z. Fan, "Growing the charging station network for electric vehicles with trajectory data analytics," in *IEEE International Conference on Data Engineering*, 2015.

[4] Y. Qin, Z. Bei, and M. Kezunovic, "Optimized operational cost reduction for an ev charging station integrated with battery energy storage and pv generation," *IEEE Transactions on Smart Grid*, no. 99, pp. 1–1, 2018.

[5] R. D. J. Tan, G. Liu and Z. Wang., "Bi-level charging station planning for integrated power distribution and transportation system," *2017 IEEE Conference on Energy Internet and Energy System Integration (EI2)*, pp. 1–5, 2017.

[6] K. Qin, Q. Zhou, T. Wu, and Y. Q. Xu, "Hotspots detection from trajectory data based on spatiotemporal data field clustering," *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLII-2/W7, pp. 1319–1325, 2017.

[7] Y. Dong, S. Qian, J. Liu, Z. Lin, and Z. Kai, "Optimal placement of charging stations for electric taxis in urban area with profit maximization," in *IEEE/ACIS International Conference on Software Engineering*, 2016.

[8] Z. Tian, T. Jung, Y. Wang, F. Zhang, L. Tu, C. Xu, C. Tian, and X. Y. Li, "Real-time charging station recommendation system for electric-vehicle taxis," *IEEE Transactions on Intelligent Transportation Systems*, vol. 17, no. 11, pp. 3098–3109, 2016.

[9] C. Luo, Y.-F. Huang, and V. Gupta, "Placement of ev charging stations–balancing benefits among multiple entities," *IEEE Transactions on Smart Grid*, no. 99, pp. 1–10, 2018.

[10] K. L. J. Wang and T. Yamamoto, "Improving electricity consumption estimation for electric vehicles based on sparse gps observations," *Energies 2017*, pp. 129–141, 2017.

[11] L. Zheng, D. Xia, X. Zhao, L. Tan, H. Li, L. Chen, and W. Liu, "Spatial–temporal travel pattern mining using massive taxi trajectory data," *Physica A Statistical Mechanics and Its Applications*, vol. 501, 2018.

[12] W. Saikong and T. Kulworawanichpong, "Comparative study of energy consumption for electric vehicles with various on-board energy storage systems," *Energy Procedia 138*, pp. 81–86, 2017.

[13] C. Fiori, K. Ahn, and H. A. Rakha, "Power-based electric vehicle energy consumption model: Model development and validation," *Applied Energy*, vol. 168, pp. 257–268, 2016.